

**CSES Problem Set****Minimizing Coins**TASK | [SUBMIT](#) | [RESULTS](#) | [STATISTICS](#) | [TESTS](#)Time limit: 1.00 s Memory limit: 512 MB

Consider a money system consisting of n coins. Each coin has a positive integer value. Your task is to produce a sum of money x using the available coins in such a way that the number of coins is minimal.

For example, if the coins are $\{1, 5, 7\}$ and the desired sum is 11, an optimal solution is $5 + 5 + 1$ which requires 3 coins.

Input

The first input line has two integers n and x : the number of coins and the desired sum of money.

The second line has n distinct integers $c_1, c_2, \dots, c_n$: the value of each coin.

Output

Print one integer: the minimum number of coins. If it is not possible to produce the desired sum, print -1 .

Constraints

- $1 \leq n \leq 100$
- $1 \leq x \leq 10^6$
- $1 \leq c_i \leq 10^6$

Example

Input:

```
3 11
1 5 7
```

Output:

```
3
```

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